

# Chat Filter Accuracy Audit

The filter searches for each banned word **anywhere in a message, including inside longer words**, and the word list is **mostly Brazilian Portuguese**. Its two-letter entries are applied to all eight languages the game ships in, so the filter blocks ordinary words in every one of them.

thank

DELIVERED

than<sup>ks</sup>

REFUSED

do<sup>cu</sup>ment

REFUSED

se<sup>cu</sup>rity

REFUSED

cu is a rude word in Portuguese. ks is a two-character entry. The filter has no rule about word boundaries, so it blocks any message containing a word that holds those letters.

15,307

MESSAGES SENT

343

WORDS CONFIRMED  
CENSORED

8,414

ENGLISH WORDS AFFECTED

16

RULES IDENTIFIED

**Content notice.** Establishing what the filter holds requires naming the terms it matches, so this report quotes profanity and slurs. They appear as evidence, not as commentary.

Black-box measurement against the live service, 11–12 September 2026. No client modification, no packet injection, no server or source access — the audit observes exactly what a player observes. All 15,307 probe records are published with this report.

## The filter matches inside longer words

The filter blocks a message wherever the letters of a banned word appear. We confirmed this by placing the letters inside a made-up word, so that a block could only be caused by those letters and not by the ordinary word that first revealed the problem.

| WE SENT THIS | WHAT HAPPENED    | WE SENT THIS | WHAT HAPPENED    | WHAT THIS SHOWS                            |
|--------------|------------------|--------------|------------------|--------------------------------------------|
| cu           | <b>blocked</b>   | ks           | <b>blocked</b>   | these letters are on the word list         |
| xxcuxx       | <b>blocked</b>   | xxksxx       | <b>blocked</b>   | the filter blocks them inside another word |
| xxcxx        | <b>delivered</b> | xxkxx        | <b>delivered</b> | the first letter alone is allowed          |
| xxuxx        | <b>delivered</b> | xxsxx        | <b>delivered</b> | the second letter alone is allowed         |
| xxucxx       | <b>delivered</b> | xxskxx       | <b>delivered</b> | the same letters reversed are allowed      |

The filter delivered 10,627 words during this audit. **None of those 10,627 words contains cu or ks.** Every word we tested that contains those letters was blocked.

## Short the filter blocks messages containing entries far more words

Ordinary English words blocked, by entry. A short entry appears by chance inside many more words than a long one does.

|       |                                                                                     |       |                             |
|-------|-------------------------------------------------------------------------------------|-------|-----------------------------|
| cu    |  | 5,016 | document, discuss, security |
| nb    |  | 956   | number, inbox, unbind       |
| meter |  | 695   | parameter, diameter         |
| anta  |  | 419   | advantage, santa, fantastic |
| ks    |  | 408   | thanks, tasks, books, works |
| pus   |  | 230   | campus, octopus, push       |
| pica  |  | 227   | typical, tropical           |
| puta  |  | 95    | reputation, computation     |

Eight further rules account for the remainder.  two-character entry  longer entry

## Coverage is uneven by language, not uniformly strict

Twenty-seven Portuguese profanity terms were tested and **all twenty-seven were refused**, down to low-frequency items like arrombado and otario. Tested identically:

|                           |                          |                          |                           |
|---------------------------|--------------------------|--------------------------|---------------------------|
| bitch<br><b>DELIVERED</b> | rape<br><b>DELIVERED</b> | slut<br><b>DELIVERED</b> | whore<br><b>DELIVERED</b> |
|---------------------------|--------------------------|--------------------------|---------------------------|

Both halves share one cause: a term list that has not been reviewed against the locales it is applied to.

## Every market, and the game's own interface text

The refusal rate was measured separately against the common vocabulary of each shipped language. It is uniform — this is not an English-only problem.



**The service censors words from its own shipped localisation files.** A player who types a word they can read on screen is told their speech is inappropriate.

## What the player experiences

The message is not delivered and the sender sees a modal stating the speech is inappropriate. Nothing indicates which word caused it, and the offending sequence is not visible as a word — so the rule cannot be learned and cannot be avoided. Repeated refusals of ordinary vocabulary read as the game being broken rather than as moderation.

## Why this costs support time

| PROPERTY                          | CONSEQUENCE                                                                       |
|-----------------------------------|-----------------------------------------------------------------------------------|
| Affects high-frequency vocabulary | thanks is among the most common words in social game chat, and it is refused      |
| Silent and unattributable         | Players cannot infer the rule, so behaviour does not adapt                        |
| Not individually actionable       | Each report names a different ordinary word produced by the same handful of rules |
| Asymmetric by market              | Indonesian shows the highest rate of the five vocabularies measured               |

## Confidence in these figures

| FIGURE                       | BASIS                                                                                                      |
|------------------------------|------------------------------------------------------------------------------------------------------------|
| 343 words confirmed censored | Measured. Each required two independent refused observations against a measured 1.2% per-probe error rate. |
| 8,414 English words affected | Derived. Confirmed rules projected across a dictionary; these specific words were not each sent.           |
| 16 rules                     | A lower bound. Entries whose letters did not occur in the 11,868 strings tested would not have surfaced.   |

## Four options. None of them blocks less.

Every option below removes false positives only — no entry is taken off the block list. They are independent and can ship in any order.

|          | APPROACH                     | TOUCHES           | RESIDUAL RISK                                | EFFORT    |
|----------|------------------------------|-------------------|----------------------------------------------|-----------|
| <b>A</b> | Ship the confirmed allowlist | Data only         | Words outside the tested set                 | Hours     |
| <b>B</b> | Add the rescue shim          | One call site     | Names and slang not in a dictionary          | 1 day     |
| <b>C</b> | Minimum-length boundary rule | Matcher condition | Long entries still match inside longer words | 1–2 days  |
| <b>D</b> | Replace the matcher          | Component swap    | None of this class                           | 1–2 weeks |

### A · Allowlist — no code change

If the filter already consults an exception list, this is a data deployment. `words-to-allow.txt` holds the 343 confirmed words, one per line, each annotated with the rule responsible.

### C · Boundary rule — best value per line

One condition addresses the cause rather than the symptoms. Long entries such as `caralho` behave exactly as they do today, so strength against substantial terms is untouched.

**Entries shorter than five characters match only as whole words.** Entries of five characters or more are unchanged.

With this rule in place, the filter still blocks a message that is only the word `cu`, and it stops blocking `document`. The filter still blocks a message that is only the word `ks`, and it stops blocking `thanks`.

## Recommended rollout

- 1 Shadow.** Evaluate the new decision alongside the current one and log where they differ, without changing what players see.
- 2 Check the direction of every difference.** All differences must be messages previously refused that would now be delivered. A message newly *refused* indicates a wiring error, not a tuning question — none of these options adds blocking.
- 3 Enable.** A and B may ship independently and in either order.

## Three lines at one call site

Call the shim where the existing filter has decided to block and knows the matched span. If that span sits inside ordinary vocabulary, suppress the block. Your matching logic, word list, severity rules and protocol are all unchanged.

```
# once, at startup
from rescue import Rescue
rescue = Rescue("allowlist-en.txt")

# inside your filter, where you were about to block:
if rescue.is_rescued(text, match_start, match_end):
    pass          # ordinary word – deliver the message
else:
    block()       # unchanged behaviour
```

**Cost:** 473,532 entries as a hash set, loaded once. Lookup is a single hash probe — not measurable against a network round trip.  
**Why it works without knowing your list:** you assert what is legitimate rather than identifying which entries are at fault, so it holds however the list changes.

## The boundary rule, language-agnostic

```
// term matched at [start, end) in the message
func accept(term, message, start, end) {
    if (len(term) >= MIN_SUBSTRING_LEN) return true;    // 5 – unchanged

    // short term: require a word boundary on both sides
    before = (start == 0)          ? null : message[start-1];
    after  = (end == len(message)) ? null : message[end];
    return !isLetter(before) && !isLetter(after);
}
```

If your matcher is a compiled automaton, tag short patterns at build time and apply the boundary test only to those, so throughput is unaffected for the rest.

## What ships in this package

| PATH                        | FORMAT                     | SIZE    | USE                      |
|-----------------------------|----------------------------|---------|--------------------------|
| findings/words-to-allow.txt | one word per line          | 343     | Option A payload         |
| findings/affected-words.txt | one word per line, by rule | 8,414   | Option A, broader set    |
| fix/allowlist-en.txt        | one word per line          | 473,532 | Option B corpus          |
| fix/rescue.py               | Python 3, no dependencies  | 138 In  | Option B shim            |
| engine/                     | Python 3, no dependencies  | ~650 In | Option D matcher         |
| vectors/golden.jsonl        | JSONL                      | 29      | Conformance for any port |

## Confirm the defect in ten seconds

Send thank in game, then thanks. The first is delivered, the second refused. This requires nothing from this package.

### Must become deliverable

| INPUT      | REFUSED BY |
|------------|------------|
| thanks     | ks         |
| document   | cu         |
| number     | nb         |
| advantage  | anta       |
| parameter  | meter      |
| reputation | puta       |

### Must stay refused

| INPUT     | NOTE                            |
|-----------|---------------------------------|
| cu        | the entry itself, as a word     |
| ks        | the entry itself, as a word     |
| puta      | short entry, still matches      |
| caralho   | long entry, rule does not apply |
| arrombado | long entry, unaffected          |

If any of these becomes deliverable the change was wired incorrectly — no option removes an entry from the block list.

```
python3 tests/test_all.py      # 55 tests
python3 tools/conformance.py   # 29 language-agnostic vectors
python3 tools/check_numbers.py # every figure here traces to the evidence
```

## 06 · METHOD & LIMITATIONS

Messages were sent through the retail client into a private party channel, one at a time, read back from chat history. Three controls govern what reached this report:

| CONTROL                                                                                 | WHY                                                                                                   |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| A sequence of letters is not reported until isolated in meaningless surrounding letters | That cucumber is refused does not identify the responsible part                                       |
| Two independent refused observations required                                           | Measured per-probe error rate is 1.2% (6 spurious in 491), so one reading is not a result             |
| An unsent probe counts as untested, not as passing                                      | All but 15 were retried to a real verdict; a false-negative audit over 250 words returned zero misses |

| LIMITATION                      | CONSEQUENCE                                                           |
|---------------------------------|-----------------------------------------------------------------------|
| Rule set is a lower bound       | There may be further entries the tested vocabulary did not reveal     |
| Dictionary counts are derived   | 8,414 is a projection; the 343 measured words are kept separately     |
| Point-in-time                   | Describes the service on 11–12 September 2026; a list edit changes it |
| Non-Latin locales under-covered | Korean, Chinese and Japanese were not swept; no conclusion is offered |

Every quantity here derives from `findings/findings.json` and is checked by `tools/check_numbers.py`, which fails if a published figure is not supported by the evidence. All 15,307 probe records: `findings/raw-logs/`.